# How to Think Different™

*Using macOS as a developer: a mini-workshop*



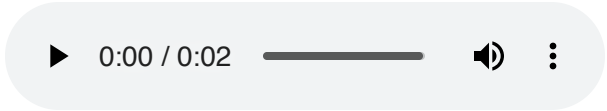
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https://github.com/Tenchi2xh/ThinkDifferent

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# I. Introduction



# What this workshop offers

I am kind of a perfectionist and like to customize everything.

What I will share with you is the result of 10 years of perfecting my setup.

The focus will be:

- A highly opinionated way of using macOS,
- Removing barriers to productivity,
- Having a neat, customized OS and tools.

# You don't have to take everything

I'm just showing you my way of working:

- Keep what you like
- Knowing about the rest will still help

# Practice makes perfect

This workshop will:

- What to avoid.
- Teach a lot of keyboard shortcuts.

**It will take some time to stick, come back to these slides.**

(And don't hesitate to contact me to ask questions!)



# My history with computers

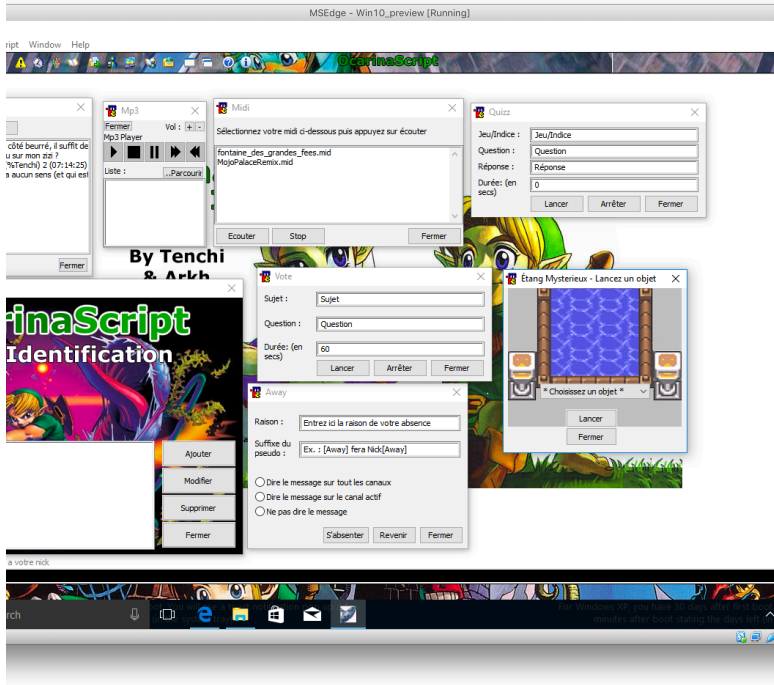


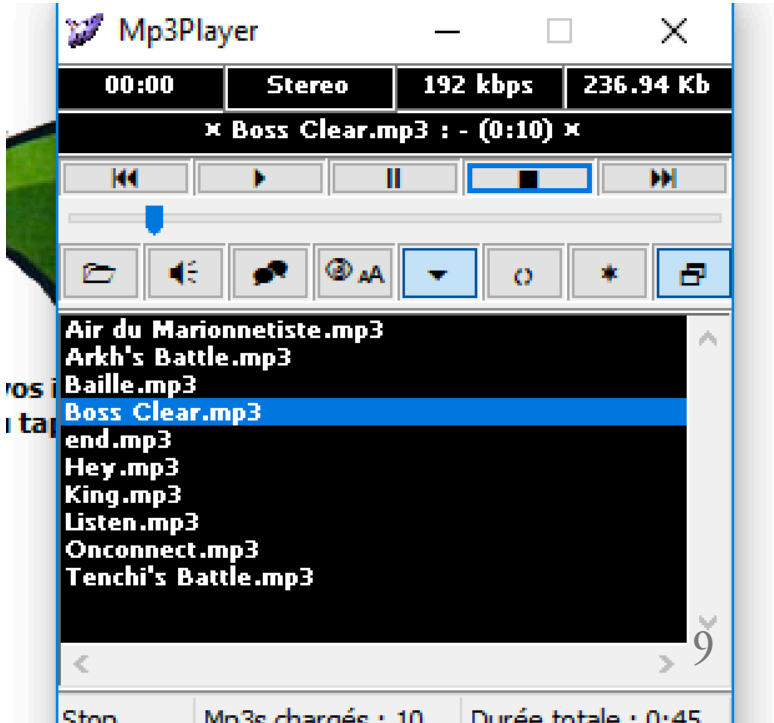

# Phase 1

- Started using a Mac at age 3-4
- Macintosh SE/30
- 1 MB of RAM (extended to 2 MB)
- 512 × 342 black and white
- Classic Mac OS:
  not suited at all for developers

# Phase 2

- Switched to windows at around age 10
- Age 11: made my own mIRC script client with ~100 users
- First Python project at age 12: Pong clone
- Random explorations of Linux many times
  - Windows is OK for dev. Experience is unstable
  - Linux is great for dev but needs a lot of care and troubleshooting








# Phase 3

- 2015: Started at Ordina's *“Scala Team”*
- Only team at Ordina with Macbooks
- Mac OS X was based on Free BSD
- With some tweaks and getting used to:
  - UNIX-like experience and easy to use
  - Great for dev! The ups of Windows and Linux without their downs.

# II. OS navigation

*It's not what it looks like! I can explain!*

# MacOS Usage overview

- Two main modes of usage:
    i. Casual: full screen apps, multiple spaces, gestures.
    ii. Power user: all in one screen, keyboard shortcuts (almost no gestures).
- Many features aimed at casual use can make things *frustrating*:
  This gives macOS a casual reputation.

→ We're going to curate features and learn some power tools.

# Basic window management

- Traffic light buttons: 
  - **Red**: close,

    **Note**: Closing the last window of an app most often doesn't quit the app.
  - **Yellow**: minimize,

    **Recommendation**: never use, forget about minimizing.
  - **Green**: full screen.

    **Recommendation**: use rarely, remember that it makes a new space.
- Double click title bar: maximize (we'll come back on this later).
- Exposé: overview of *all windows* or *current application*.

# A note about closing windows

- On Windows, closing an app's main window exits that app.
- On macOS, the app keeps running:
  - Focusing that app shows no window, but the menu bar has its name and menu items.
  - This results in a lot of zombie apps that take ram and do nothing.
  - Only by quitting the app with ⌘ + Q or using the menu actually quits the app.

Don't close windows, just use ⌘ + Q.

# A note about minimization

- On Windows, minimizing a window just hides it.
- On macOS, minimizing a window puts it in a special prison cell:
  - It disappears from alt-tab,
  - It disappears from Exposé,
  - It literally goes to prison, in the tiny tiny dock.

Don't minimize windows, they will be sad, and you too.

# Spaces

- Each space has its own windows.
  - Alt tabbing between apps that are in multiple spaces can be confusing.
- Each full screen *window* is in its own space (even a full screen YouTube video).
- Exposé part 2: move windows to other spaces.

**Recommendation**: never use spaces, unless you need one app in full screen (for a presentation).

# A note about full screen

- On Windows, clicking the “maximize” icon (here the green ball  ) just makes the window use as much space as possible on the screen.
- On macOS, it puts the application in full screen. Full screen apps go to their own dedicated space:
  - It disappears from alt-tab,
  - It disappears from that app's Exposé (but goes to the top of the Exposé screen).

Don't full screen windows, unless you want to look neater during a presentation.

# A note about the traffic light buttons 

**Recommendation**: avoid using.

# Spotlight

One of the most useful shortcuts: `⌘` + `Space`.

Use it for everything:

- Launching apps (launching same app switches focus to it),
- Searching files,
- Quick web search.

**Recommendation**: use it all the time.



# The desktop

**Recommendation**: never use.

# The dock

- Basically just a list of app shortcuts.
- Not like the Windows taskbar, each icon is one application, not window.
- “Minimized” windows go down there on the right: this is not useful at all, avoid minimizing.

**Recommendation**: can be used for quick app shortcuts, but just use Spotlight instead.

# Finder

The “Explorer” of macOS. It's better because you don't explore, you find.

Worthy of note:

- You can't exit that the Finder. It runs forever and `⌘` + `Q` does nothing.
- Supports tabs (`⌘` + `T`).
- Dragging files moves, hold `⌥` to copy.
- Use `Space` to preview things (more on that on the *Preview* slide).

# III. Keyboard shortcuts

*A frankly unreasonable amount of shortcuts.*

# Why keyboard shortcuts

Our focus today:

- Navigating the OS without clicking around,
- Editing text efficiently,
- Being in control and overall productivity.

It will feel like a lot, but I didn't look up any of these: it's really the list of stuff I use daily.

**Note**: The following slides apply OS-wide.

# The cast of characters

In order of awesomeness (subjective).



# Command

Symbolized as ⌘

Most main actions (or *commands* see what they did there) of an app use ⌘ (and not the *Control* key like in Windows).

**Fun facts**:

- Chosen by Susan Kare, designer of the icons in classic Mac OS and the Solitaire cards in Windows 3.1.
- In Scandinavia: denotes a place of cultural interest.

# Control

Symbolized as ^

Macs have both *Command* and *Control*,
and it makes them keyboard-shortcut monsters.

- Windows and Linux barely use their *Meta* key, and its usage is for global shortcuts.
- On Mac, *Command* is completely application-oriented, tripling keyboard shortcuts:
  - ⌘ + …
  - ^ + …
  - ^ + ⌘ + …

**Fun fact**: the ^ is used on road signs to indicate roofed infrastructure.






# Option

Symbolized as ⌥

*Option* is the macOS name for *Alt*:

- Used less for keyboard combinations (but still used).
- ⌥ + any letter makes symbols.
- Rarely used for shortcut combos.

**Fun fact**: I'm *really* struggling to keep the road sign theme going on.

# Fn

- Simulate missing keys (Page up, home, etc.).
- Used for F-Keys.
- Switch keyboard layouts.




*Yep I give up, best I could find.*

**Fun fact**: *It's a weapons factory in Belgium.*

# The shortcuts

In order of whatever (subjective).

# Keyboard window management

| Shortcut | Action |
| --- | --- |
| ⌘ + Tab | Tabbing through **applications** ( ⇧ : reverse). |
| ⌘ + < / > | Tabbing through **windows**. |
| ⌘ + W | Close window. |
| ⌘ + ⇧ + W | Close all windows in current app. |
| ⌘ + Q | Quit application. Recommended: disable in some important apps. |
| ^ + ⌘ + F | Toggle full screen (same as green button). |

# Clipboard

- Keyboard: `⌘` + `X` / `C` / `V`: Cut / Copy / Paste.

  **Note**: since copying is bound to ⌘, this means that `^` + `C` is still available!

- Terminal: `pbcopy` and `pbpaste` commands, very useful:

```
cat slide.md | pbcopy  # copies the contents of slide.md into clipboard
pbpaste > new_slides.md  # pastes the contents of the clipboard into new file
echo "Hello, $(pbpaste), bye bye" > wrapped_slides.md  # Can be used like any command
```

# Classic UNIX keyboard shortcuts

MacOS inherits some UNIX terminal keyboard shortcuts **OS-wide**.

That's right, not just in terminals, but **everywhere you can type text**.

| Shortcut | Action |
| --- | --- |
| ^ + A / E | Jump to beginning/end of line. |
| ^ + T | Transpose letters. |
| ^ + B / F | Move one character left/right. |

# Moving around text

| Shortcut | Action |
| --- | --- |
| Fn + ↑ / ↓ | Page up/down. |
| Fn + ← / → | Home/end. |
| Fn + ⌫ | Erase to the right (like Del). |
| ⌥ + ← / → | Move cursor a whole word left/right. Combine with ⇧ for selection. |
| ⌘ + ← / → | Move to beginning/end of physical. line<br>(behavior slightly different than ^ + A / E). |

# Searching

| Shortcut | Action |
| --- | --- |
| ⌘ + F | Start search. |
| ⌘ + G | Next result. |
| ⌘ + ⇧ + G | Previous result. |

**Note**: Shift ⇧ often inverses the effect of a shortcut (or expands its scope, like "Close all windows").

# Screenshots

Prepare for some carpal tunnel syndrome:

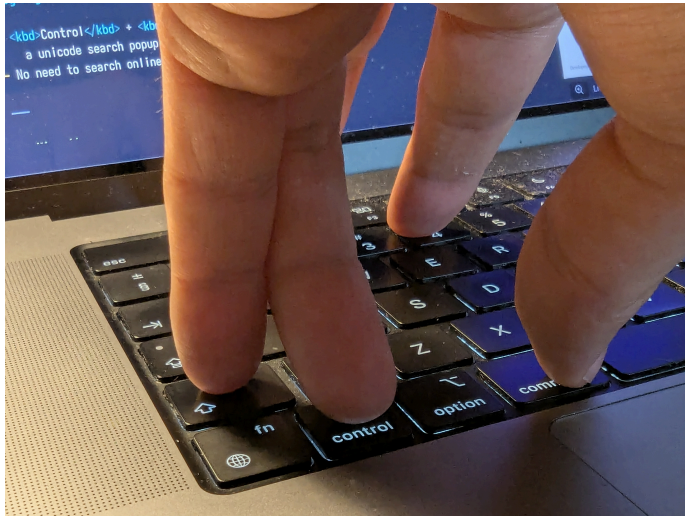
| Shortcut | Action |
| --- | --- |
| ⌘ + ⇧ + 3 | Capture whole screen, save to configured folder. |
| ^ + ⌘ + ⇧ + 3 | Capture whole screen, save to clipboard. |
| ⌘ + ⇧ + 4 | Capture selection (drag mouse), save to configured folder. |
| ^ + ⌘ + ⇧ + 4 | Capture selection (drag mouse), save to clipboard. |
| ⌘ + ⇧ + 5 | Screen recording tool. |
| ^ + ⌘ + Esc | Stop recording. |

# Capture selection to clipboard

You're probably thinking: *"WTF."*

1. First place your thumb on ⌘,
2. Then place your pinky on ⇧,
3. Then place your ring finger on ^,
4. Finally, flatten your hand and reach for 4 with your index finger.
5. You can now release the keys!
6. Drag a box with your mouse and release.

**Pro tip**: while dragging the box, keep holding mouse click and hold Space to reposition the box.



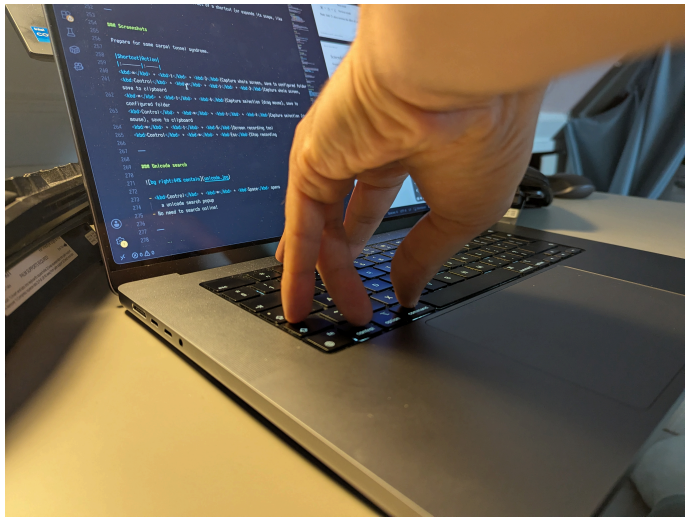



Figure 38.a and 38.b: How to RSI



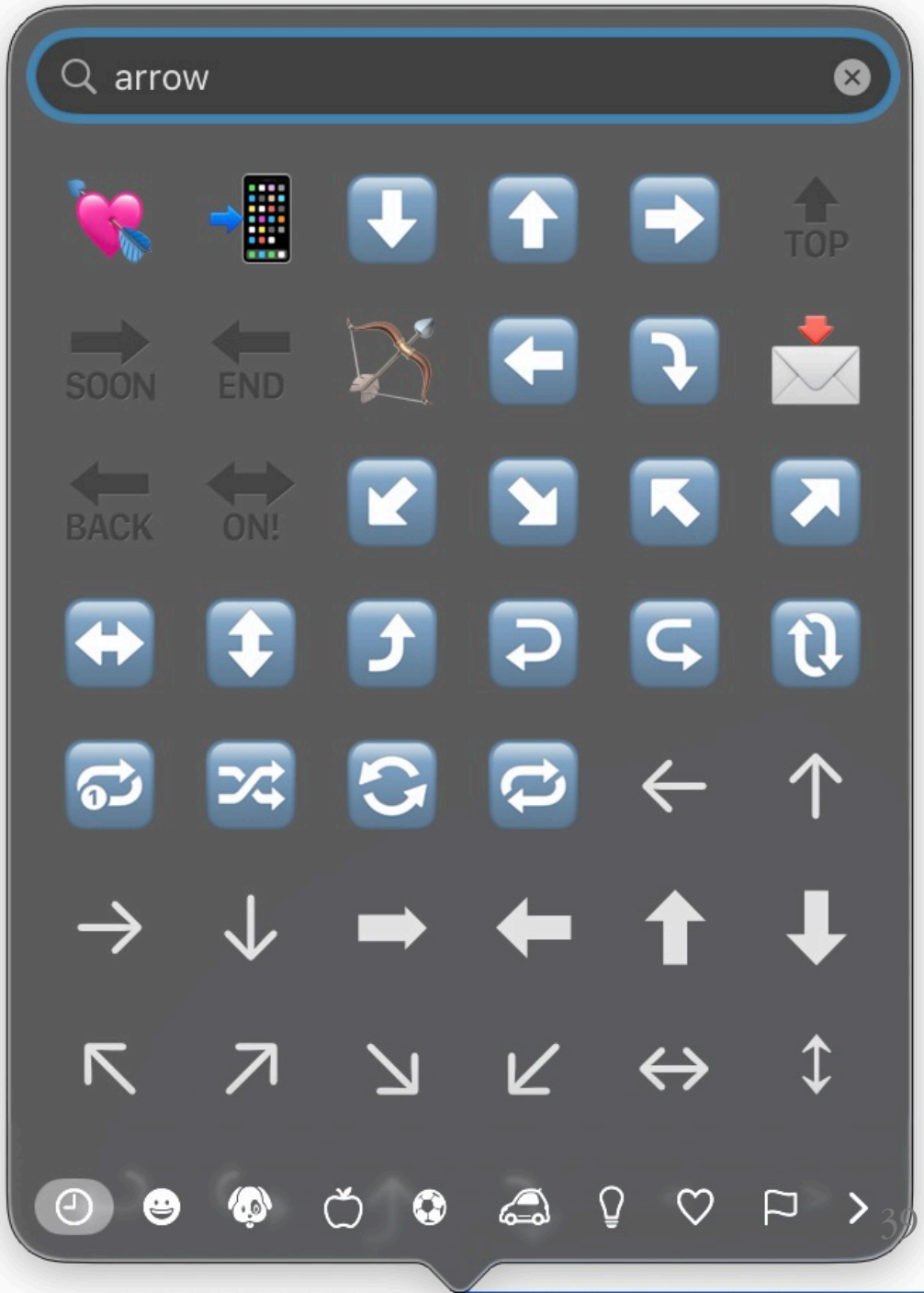

# Window mode

In mode 4 (drag a box thingy), press `Space` to switch to window capture mode.

- Captures a single window,
- With PNG transparency and shadows,
- Isolated from anything in the background.

# Unicode search

- ^ + ⌘ + Space opens a Unicode search popup.
- No need to search online!



# Miscellaneous

- Long hold a vowel to open a popup with accented variants.
- ⌥ + any letter makes symbols.
- ⇧ + ⌥ + any letter makes even more symbols.
  - For example, to make “Pretty quotes”: (⇧ +) ⌥ + [
- Application settings standard shortcut is ⌘ + ,
  - If you try ⌘ + p like on Windows, that's the *Print* command instead.

# Gestures

| Gesture | Action |
| --- | --- |
| Three fingers up | Show all windows across all apps (and spaces). |
| Three fingers down | Show all windows in current app. |
| Swipe three fingers to the left/right | Switch spaces (or switch to full screen app). |
| Place four fingers down then spread them | Show desktop (🤮). |
| Drag two fingers in any direction in a scrollable area | Scroll (direction depends on “natural direction” setting). |
| Drag two fingers to the left/right | Navigate to previous/next page (in a browser). |
| Drag two fingers from the right, outside of the trackpad | Show notifications. |

# IV. Pitfalls

*Tiny chapter to make it a nice three-column table of contents.*

# Common issues

If you can't find your app:

- You might have minimized it with ⌘ + M or the 🟡 button.
  Look at your dock in the bottom right area.
- You might have full screened it with ^ + ⌘ + F or the 🟢 button.
  Look at your spaces by swiping three fingers around.

You have a lot of apps running for no reason. Tabbing to them shows nothing:

- You closed all the windows of that app using ⌘ + W or the 🔴 button.
  Closing windows does not quit apps.
  Quit the app with ⌘ + Q or click the main menu of that app (just right of the Apple logo) and choose “Quit”

# Advice

- Don't minimize, and only full-screen for videos or presentations.
- Keep all your windows in one space. To find your windows, do either:
  - A combination of `⌘` + `Tab` and `⌘` + `<` / `>`.
  - A combination of `⌘` + `Tab` and the three fingers down gesture.
  - Three fingers up gesture (if you have a very recognizable window).
- When you intend to quit an app, don't use the 🔴 button. Quit the app instead.

# V. Third-party tools and applications

*The rest of the batteries.*

# Preamble

Everything after these slides will be automatically installed during the last chapter.

These slides are for demos, and later for reference.

# Homebrew

| | |
|---|---|
| **Description** | De facto standard package manager for macOS, maintained by the community. |
| **Model** | FOSS |
| **Link** | https://brew.sh |
| **Install** | `/bin/bash -c "$(curl -fsSL https://raw.githubusercontent.com/Homebrew/install/HEAD/install.sh)"` |

**Notes**:

- Main function: install packages and dependencies
- Secondary function: install GUI applications (called “casks”)

# iTerm2

| | |
|---|---|
| **Description** | `Terminal.app` sucks. Use this instead. |
| **Model** | FOSS |
| **Link** | https://iterm2.com |
| **Install** | `brew install --cask iterm2` |

**Good alternatives**:

- Ghostty
- kitty
- Alacritty

- Best feature: global hotkey. I set it to ⌘ + §.
- Second best-feature: automatic copying.

Useful shortcuts:

- ⌘ + T: New tab.
- ⌘ + K: Clear.
- ⌘ + D: Split vertical.
- ⌘ + ⇧ + D: Split horizontal.
- ^ + D: Close tab/panel (standard UNIX for logging out).

# Sublime Text

| | |
|---|---|
| **Description** | Lean, fast and customizable text editor. |
| **Model** | Paid, with eternal free trial |
| **Link** | https://www.sublimetext.com |
| **Install** | `brew install --cask sublime-text` |

**Good alternatives**:

- neovim for vim people.
- Zed

*Built-in TextEdit app not recommended.*

- Not a replacement to IDEs, just a good text editor.
- Nice as a default for opening text-based files.
- Many plugins and themes.
- Very easy to hack via Python.
- Opens instantly.
- Persists all windows and tabs.
- Very good multi-line editing (invented the concept!).

# VI. Third-party extensions

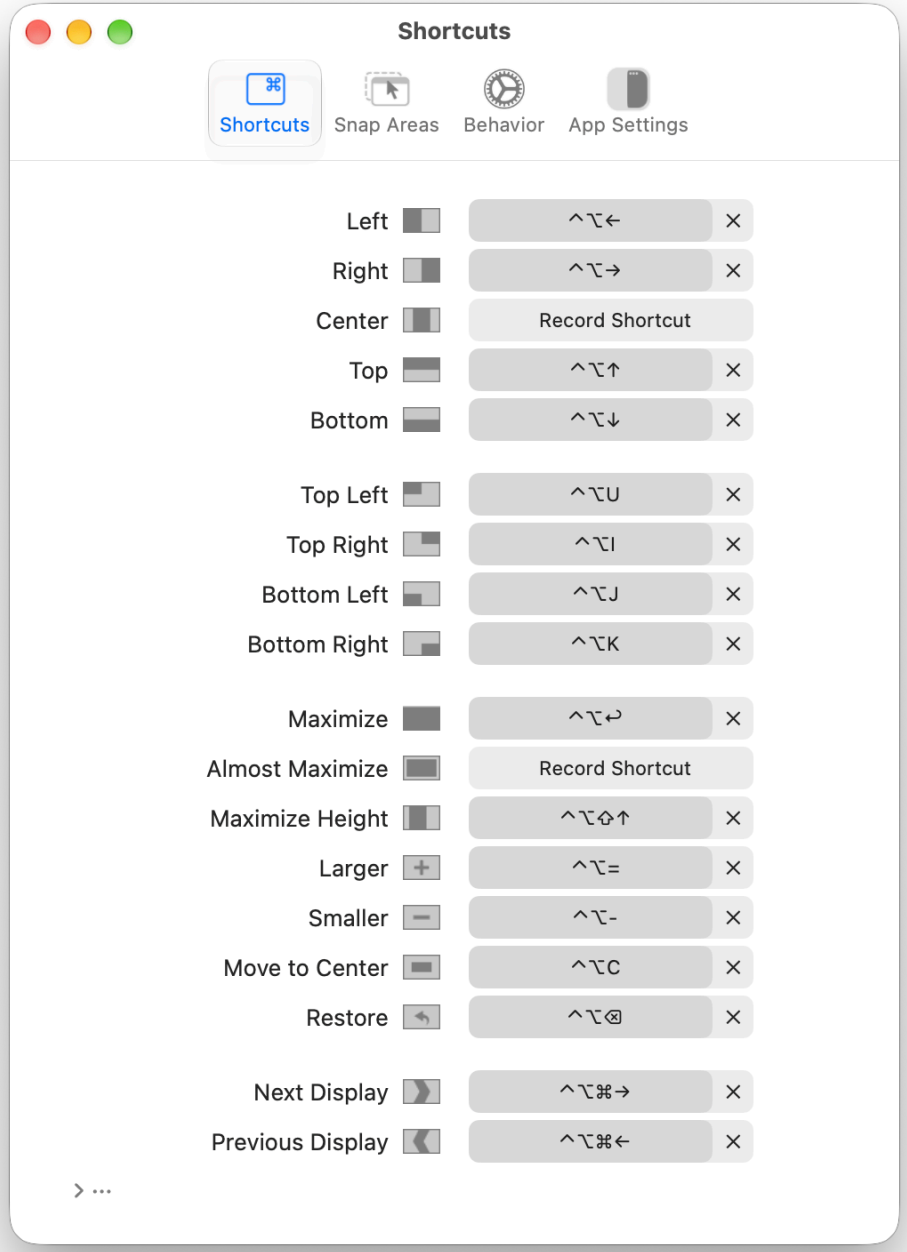
*Unofficial tools to customize the OS more, or add missing functionalities.*

# Rectangle

| | |
|---|---|
| **Description** | Window management app |
| **Model** | FOSS |
| **Link** | https://github.com/rxhanson/Rectangle |
| **Install** | `brew install --cask rectangle` |

**Notes**:

- MacOS has its own shortcuts, but the animations are long, and the shortcuts not customizable or natural.
- My favorite shortcut prefix for this app is ⌥ + ⌘.
- I said we'll come back to it later: to maximize, I don't double click a title bar, I do ⌥ + ⌘ + F.

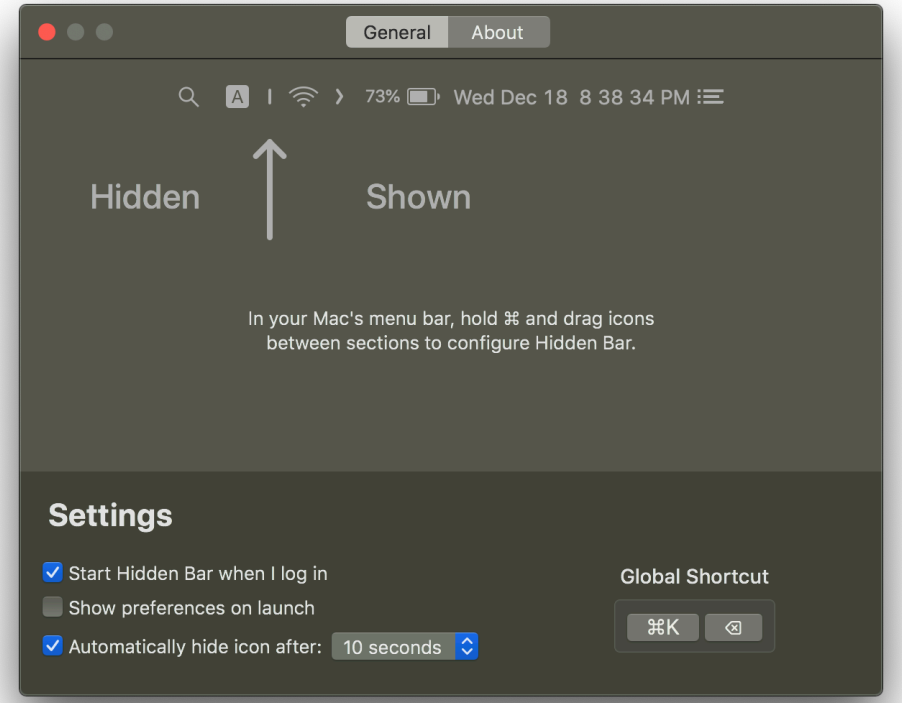


# Hidden Bar

| | |
|---|---|
| **Description** | Hides your excess menu bar icons |
| **Model** | FOSS |
| **Link** | https://github.com/dwarvesf/hidden |
| **Install** | `brew install --cask hiddenbar` |

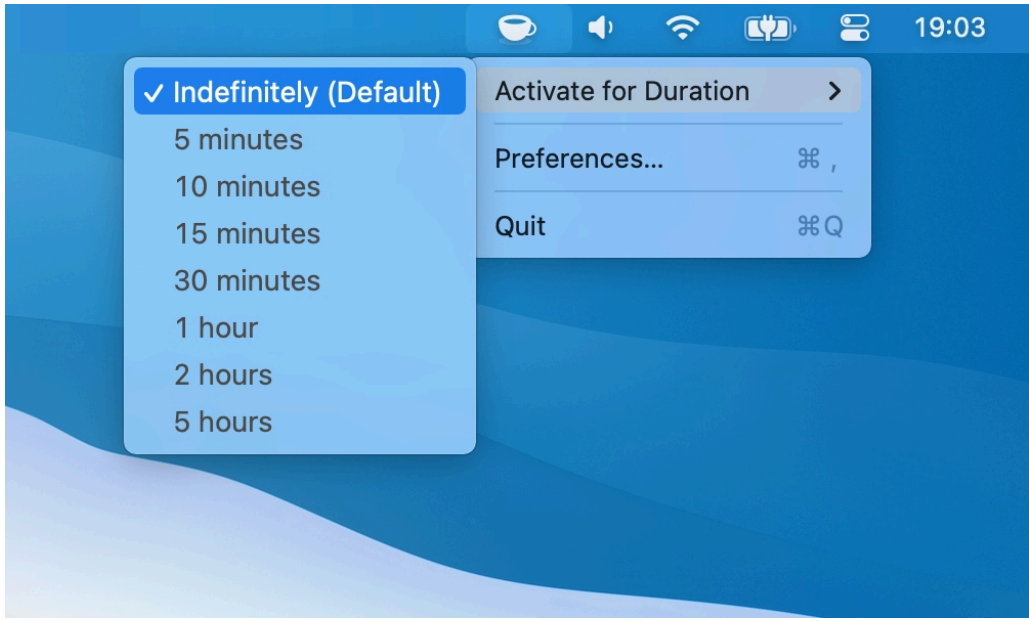
**Notes**:

- Can't believe this is not in standard macOS.
- Click the arrow to expand.
- Right click the arrow to show super-hidden icons.



# Keeping You Awake

| | |
|---|---|
| **Description** | Prevents your Macbook from sleeping |
| **Model** | FOSS |
| **Link** | https://github.com/newmarcel/KeepingYouAwake |
| **Install** | `brew install --cask keepingyouawake` |




**Notes**:

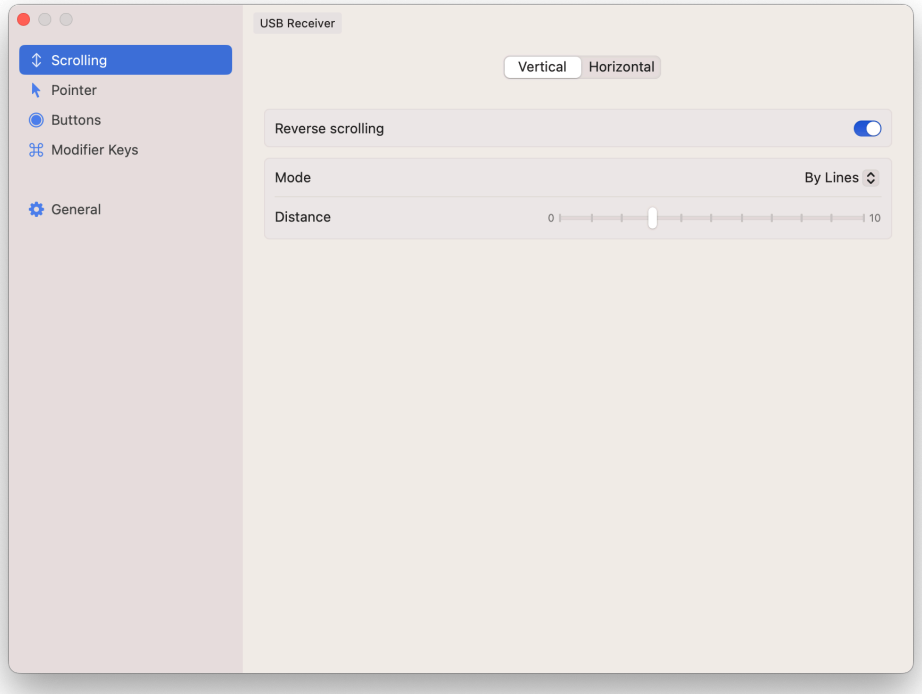
- Quick toggle for letting your Mac sleep on idle or not.

# Linear Mouse

| | |
|---|---|
| **Description** | Disables acceleration on physical scroll wheels |
| **Model** | FOSS |
| **Link** | https://github.com/linearmouse/linearmouse |
| **Install** | `brew install --cask linearmouse` |

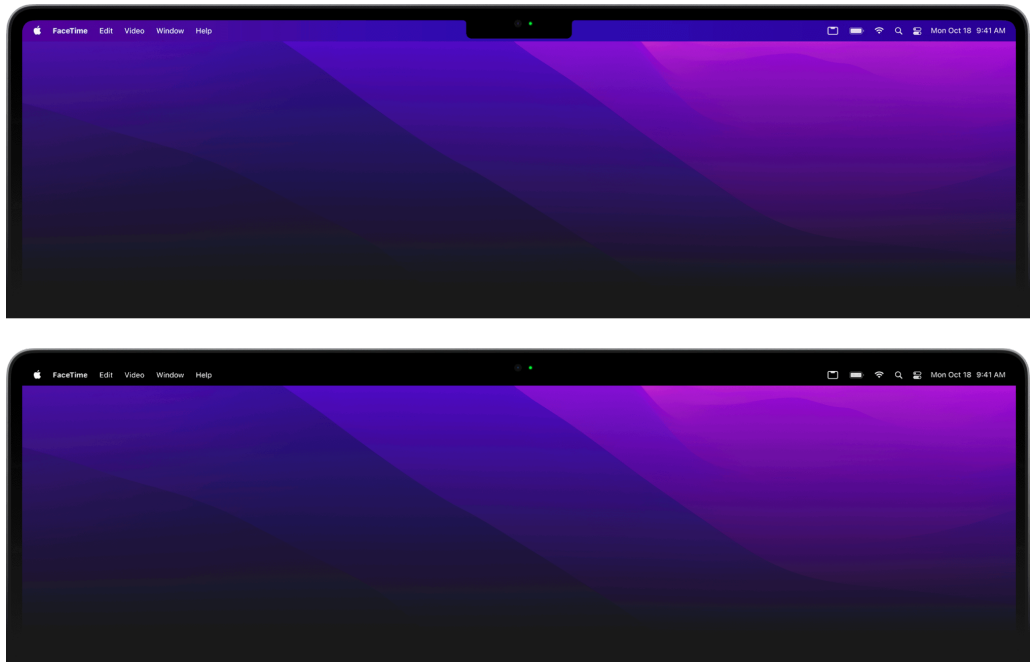
**Notes**:

- On macOS, the faster you scroll, the more it scrolls.
  If you scroll just one tick, it will literally move one pixel.
- With this app:
  One mousewheel tick = a fixed number of lines.



# Top Notch

| | |
|---|---|
| **Description** | Hides the stupid notch |
| **Model** | Free, but not open source |
| **Link** | https://topnotch.app |
| **Install** | `brew install --cask topnotch` |




**Notes**:

- To make laptops uslessly sexier, Apple reduced the bezel by adding a thick ugly notch.
- This modifies your wallpaper and adds a black bar.
- Not only you can't see the notch, but it actually looks cooler.
- Sometimes it forgets to do it, just click the menu bar icon.

# SwitchResX

| | |
|---|---|
| **Description** | Switch any monitor to any resolution |
| **Model** | Paid: 14€ lifetime license |
| **Link** | https://www.madrau.com |
| **Install** | `brew install --cask switchresx` |

**Notes**:

- Apple thinks resolutions are scary for casuals, so they just give DPI scaling options.
- With this you can control exactly what resolution, scaling and refresh rate for anything plugged into your Mac.
- Very useful for presentations.

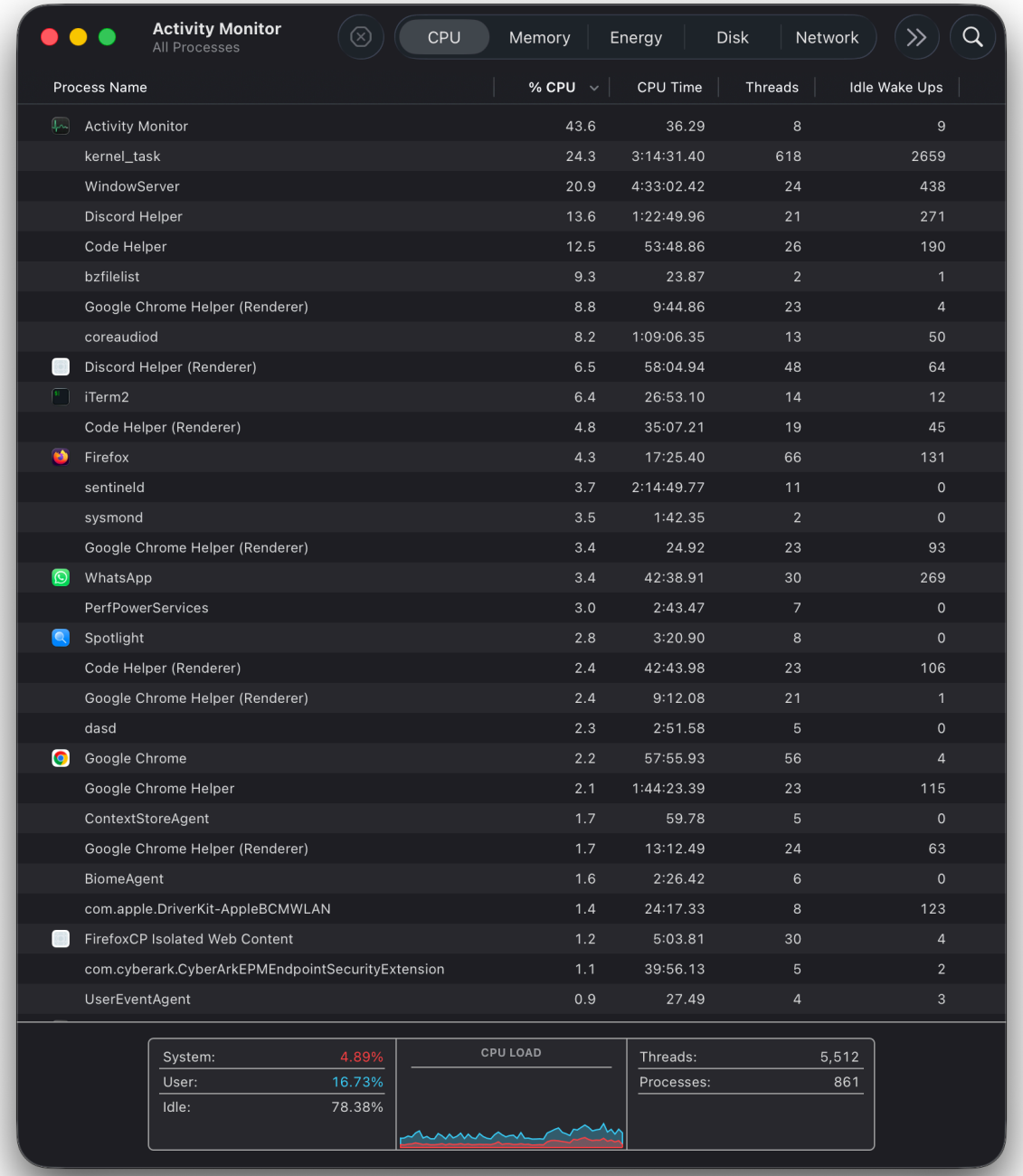


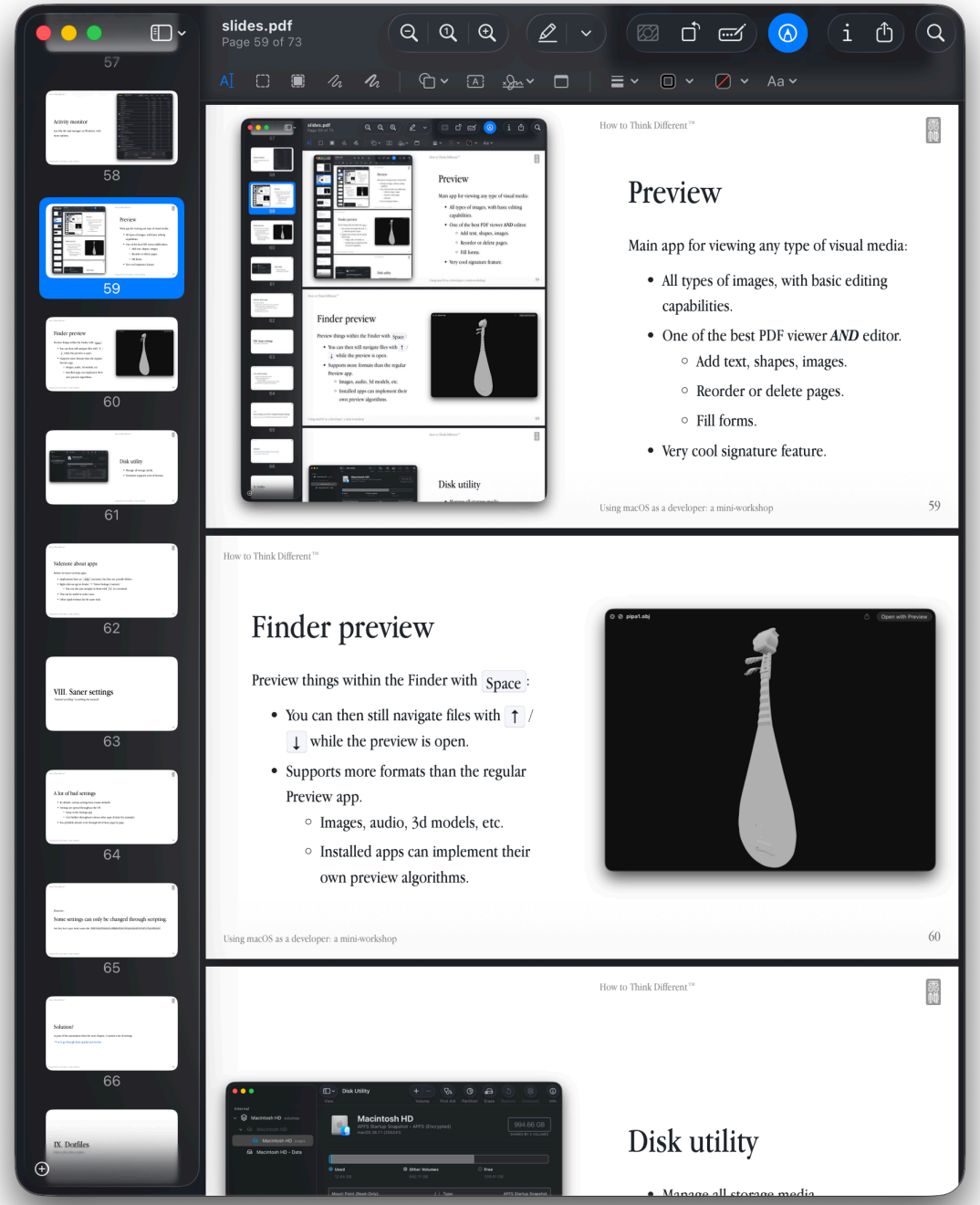
# VII. Useful built-in applications

*Comes with some batteries included.*

# Activity monitor

Just like the task manager on Windows, with more options.








# Preview

Main app for viewing any type of visual media:

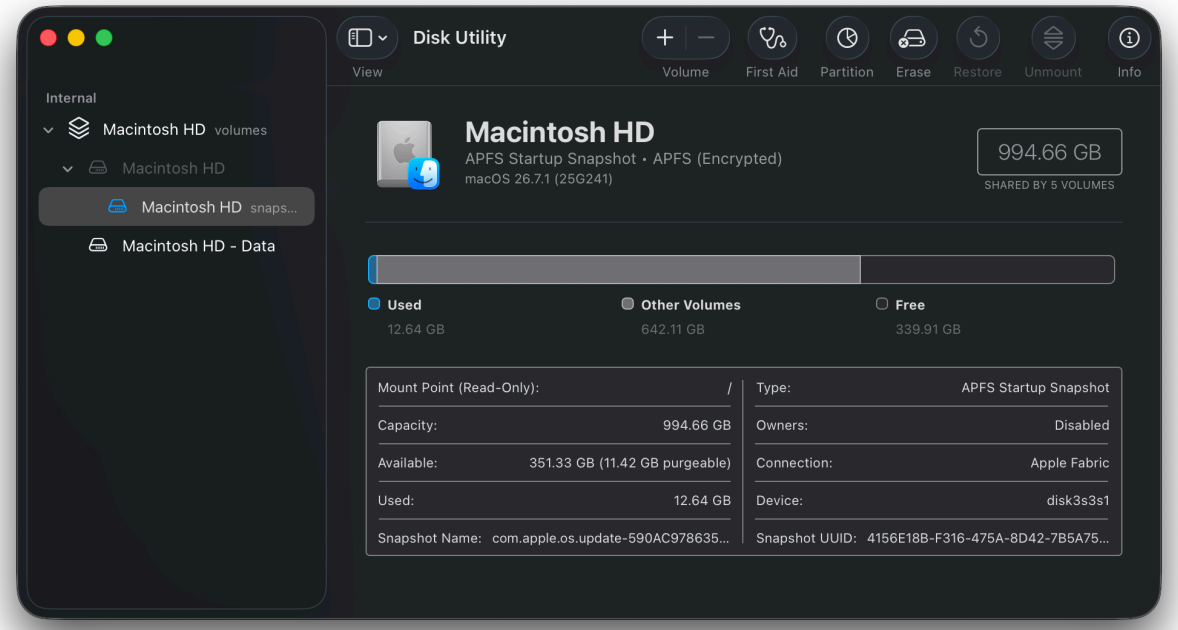
- All types of images, with basic editing capabilities.
- One of the best PDF viewer ***AND*** editor.
  - Add text, shapes, images.
  - Reorder or delete pages.
  - Fill forms.
- Very cool signature feature.

# Finder preview

Preview things within the Finder with `Space`:

- You can then still navigate files with `↑` / `↓` while the preview is open.
- Supports more formats than the regular Preview app.
  - Images, audio, 3d models, etc.
  - Installed apps can implement their own preview algorithms.






# Disk utility

- Manage all storage media.
- Formatter supports a lot of formats.

# Sidenote about apps

Before we move on from apps:

- Applications have an `.app` extension, but they are actually folders.
- Right click an app in Finder → “Show Package Contents”.
  - You can also just navigate to them with `ls` in a terminal.
- This can be useful in some cases.
- Other Apple formats do the same trick.

# VIII. Saner settings

*“Natural scrolling” is nothing but natural.*

# A lot of bad settings

- By default, various settings have insane defaults.
- Settings are spread throughout the OS:
  - Some in the Settings app
  - A lot hidden throughout various other apps (Finder for example)
- You probably already went through all of those page by page.

*However...*

# Some settings can only be changed through scripting.

And they have super funky names like `NSGlobalDomain.NSNavPanelExpandedStateForSaveMode2`

# Solution?

As part of the automation from the next chapter, I curated a list of settings.

→ Let's go through them quickly just for fun.

# IX. Dotfiles

*Hide yo files, hide yo folders.*

# What are dotfiles?

- On UNIX-like environments, hidden files or folders.
- Most tools will pollute your `~/` full of dotfiles.
- Configuring a whole new computer can take a week.
  - People have started version controlling their home folder dotfiles.
  - Even better: they made tools to manage and apply their version-controlled dotfiles.
  - It's like CI for your own computer.

# Chezmoi

- Tool for applying dotfiles easily.
- Every file/script is templatable.
- Supports various triggers for executing scripts.
- Support for many password managers.
- Explicit name-based permissions: private files, executables, etc.
- Dotfiles made by Chezmoi are symlinks

  → **Change your settings in an app, the change is made directly in your repository**.
- And many other features I don't use.

# My dotfiles

- All my configs, settings, custom scripts, in one repo: https://github.com/Tenchi2xh/dotfiles
- What it does:
  - Installs apps, extensions and CLI tools,
  - Setup custom file associations,
  - Configure macOS with custom settings,
  - Configure `zsh` with custom theme, animations and plugins,
  - Configure some third-party tools,
  - Installs and uses my own customized terminal font,
  - Optional: organize the dock and set up custom icons.
- Very opinionated and themed around my branding, but easily customizable.

# Requirements

- Install Homebrew:

```
/bin/bash -c "$(curl -fsSL https://raw.githubusercontent.com/Homebrew/install/HEAD/install.sh)"
```

- Install Chezmoi: `brew install chezmoi`.
- If using Bitwarden: `brew install bitwarden-cli`.
- Create a `SSH Key` entry in bitwarden, add your private SSH key in the notes field.
  → If not desired, delete `dotfiles/private_dot_ssh/private_id_rsa.tmpl`
- Create an `Auth tokens` entry in bitwarden, add a custom field `HOMEBREW_GITHUB_API_TOKEN`.
  → If not desired, delete the line `export HOMEBREW_GITHUB_API_TOKEN=...` in `dotfiles/dotfiles/dot_zprofile.tmpl`.

# How to use them

- Fork my repo https://github.com/Tenchi2xh/dotfiles.
- Leave a ★.
- In the `config/` folder, remove or edit what you don't need from the YAMLs (leave empty lists when needed).
- In the `scripts/` folder, remove scripts you don't care about (for example Dock customization, custom icons).
- In the `dotfiles/` folder, remove config files that don't apply to you (also replace the public SSH key).
- `commit` and `push`.
- If using Bitwarden: `bw login` and `bw unlock`.
- Apply: `chezmoi init --apply [YOUR_GITHUB_USERNAME]`



# Peace.